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Pulchowk, Nepal

A

Progress on

SoC–Based Inverter Control Strategy for Grid-Connected Battery

Energy Storage in AC Microgrid

Submitted by

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Submitted to

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Pulchowk, Nepal

August, 2026

Acknowledgment

I would like to appreciate **Er. Jeetendra Chaudhary**, HoD, Department of Electrical Engineering . The suggestion and task assigned by department head has helped a lot in progressing the project work. We would like to express our gratitude to Pulchowk Campus for the opportunity of undertaking the Second Year Project.

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List of Abbreviations

AC:	Alternating Current
BESS:	Battery Energy Storage System
CIGRE:	International Council on Large Electric Systems
DC:	Direct Current
DOD:	Depth of Discharge
EMS:	Energy Management System
EV:	Electric Vehicles
GTI:	Grid-Tied Inverter
MG:	Microgrid
MW:	Mega Watts
P-Q:	Active Power-Reactive Power (control)
P-V:	Active Power-Voltage (control)
PCC:	Point of Common Coupling
PI:	Proportional-Integral
PLL:	Phase-Locked Loop
PV:	Photovoltaic
PWM:	Pulse Width Modulation
RES:	Renewable Energy Source
SOC:	State of Charge
SOH:	State of Health
SRF:	Synchronous Reference Frame
WT:	Wind Turbine

Chapter 1

Introduction

1.1 Background

The drop in the price of the battery has caused the spike in its use. This has made a favorable market for the battery and energy storage system. In recent years, there has also been an increase in the integration of intermittent sources of energy such as PV and wind. This has made it much more crucial for the application of the battery to reduce the effect of intermittent sources. Application of the battery also comes at a cost. Despite the drop in its price, the use case and method of integration to the system is crucial. Currently, batteries are implemented in sectors where there is a high cost of interruptions. Along with that, BESS are now being integrated into the grid in the form of energy arbitrage. BESS can help to provide ancillary services to the grid and help to maintain grid stability. Due to its importance, batteries are required to be implemented in such a way that they can have a high lifespan. Batteries can drain out, have chemical imbalance, and have cell-level issues if the batteries are not handled properly. Lifespan of the batteries is determined on the basis of the use case and application. For a microgrid operation, SOC is the most important metric as it helps to determine the usable energy stored in the battery storage.

1.2 Statement of Problem

The models which are currently implemented in the BESS-based inverter only account for the load and its variation for the droop regulation. The grid-tied inverter (GTI) based inverter only follows the droop-regulated system. This hampers the system which contains the costly battery. Utilizing the BESS to its maximum capacity comes with its degradation of the lifespan. Just utilizing the BESS for maximum benefit cannot happen when it is solely used as a backup system. Effective power sharing based on the SOC and its adaptation can help in the longevity of the battery and the overall system. However, this can reduce the power output from the BESS device, but it can also be mitigated by the implementation of a

smart EMS system.

1.3 Objective of Study

Main Objective

- To create a droop-regulated BESS-based system with an SOC-based inverter system for an AC microgrid.

Specific Objective

- To develop a control logic to be implemented in an inverter control
- To identify the parameters of the inverter based on the design specification
- To validate the droop control in the simulated environment

Chapter 2

Literature Review

Droop control is the most widely used method for autonomous, communication-free power sharing among inverter-interfaced sources in a microgrid, since it mimics the frequency- and voltage-dependent behaviour of synchronous generators [2]. Conventional droop schemes regulate active and reactive power output based on deviations of grid frequency and voltage from their nominal values, but a large fraction of the existing literature does not explicitly account for the state of charge (SOC) of the battery energy storage system (BESS) supplying the inverter, which can result in inefficient power management, accelerated battery degradation, and suboptimal system performance [3].

A number of studies have attempted to incorporate SOC into the droop framework, but predominantly for islanded or DC microgrids. Wu et al. [4] proposed a droop strategy to achieve SOC balance among multiple energy storage units of varying capacity in an islanded AC microgrid. Su et al. [5] combined droop control with DC bus signalling to coordinate BESS operation in a DC microgrid without requiring a communication link, while Hoang and Lee [6] presented a voltage-restoration method that achieves accurate power sharing and SOC balancing among distributed battery units. Zeng et al. [7] and Belal et al. [8] proposed decentralised control strategies that adjust droop coefficients according to SOC and battery capacity to achieve SOC balance and bus-voltage stability in islanded DC microgrids, and Wang et al. [9] introduced a virtual-battery-based droop method that dynamically adjusts the droop coefficient and reference output voltage of the storage system based on its charge state.

Comparatively fewer works address SOC-aware control explicitly for AC microgrids, and most of these focus only on achieving SOC balance among multiple units rather than on optimising the energy utilisation of an individual BESS-fed inverter while it is exporting active and reactive power to the grid [10, 11]. Al-labadi et al. [1] address this gap directly: they propose an SOC-based droop control strategy for a single grid-connected, BESS-fed inverter operating within a modified CIGRE low-voltage microgrid benchmark. Their scheme augments a conventional synchronous-reference-frame (SRF) current controller with an SOC-voltage droop

loop, so that the reference d -axis current supplied to the inner current controller is adjusted according to the deviation between the measured and reference SOC. The strategy is shown, through MATLAB/Simulink simulation, to reduce the inverter's active-power reference smoothly as the battery discharges, to disconnect the inverter safely when the SOC nears its lower protection limit, and to maintain stable operation under step changes in the reactive-power reference. The authors further compare this with a power–voltage (P–V) droop variant, which regulates the terminal voltage directly rather than reactive power, and report improved voltage regulation performance under varying SOC conditions.

This proposal builds directly on the work of Allabadi et al. [1], using their control formulation and benchmark system as the basis for an independent MATLAB/Simulink implementation, in order to validate the reported behaviour and to serve as a foundation for further exploration of SOC-based control strategies for BESS-fed inverters in AC microgrids.

Chapter 3

Methodology

The methodology follows the control strategy and benchmark system reported by Allabadi et al. [1], implemented independently in MATLAB/Simulink (Sim-scape Electrical, Specialized Power Systems library) to reproduce and validate the reported results.

3.1 Microgrid Modeling

A modified version of the CIGRE low-voltage microgrid (MG) benchmark is used as the test system. The network consists of a 0.4 kV, 50 Hz main feeder supplying six laterals with defined line resistance and inductance, four distributed loads (with a combined rating of 202 kVA), two photovoltaic (PV) sources, one wind turbine (WT), and two BESS units, each rated at 30 kW, connected through grid-following inverters. The line and load parameters are taken directly from the benchmark to allow like-for-like comparison of results.

3.2 BESS and Power Sharing Modeling

Each BESS-fed inverter is controlled using a closed-loop current controller implemented in the synchronous reference frame (SRF). Measured three-phase voltages and currents are transformed to the dq -frame using Clarke and Park transformations referenced to a phase-locked loop (PLL). Measured active and reactive power (P_m , Q_m) are compared against dispatched references (P_{ref} , Q_{ref}) through an outer PI controller to generate reference dq -currents. A second, inner PI controller compares these reference currents against the measured dq -currents to generate the reference voltage, which is transformed back to the abc -frame and passed to the PWM generator.

The SOC-based droop loop is incorporated as follows. The deviation between the reference SOC and the measured SOC of the BESS ($\Delta SOC = SOC_{ref} - SOC_m$) is scaled by an SOC-voltage droop coefficient k_{SOC-V} to obtain a voltage correction term ΔV , which is in turn scaled by a coefficient k_{V-I} to produce a

correction current ΔI . This correction is added to the reference d -axis current before it is passed to the inner current-control PI loop, so that the active power delivered by the inverter is automatically reduced as the battery discharges, and increased toward the nominal set point as the SOC approaches its reference value. A lower SOC protection limit (35%) triggers disconnection of the inverter to prevent over-discharge, and an upper limit (85%) is used to hold the nominal power reference constant.

3.3 Multi-Layer Power System Architecture

The overall control architecture is organised in three layers: (i) an upper dispatch layer, which sets P_{ref} , Q_{ref} , and SOC_{ref} for each BESS unit; (ii) a middle SOC-droop layer, which adjusts the d -axis current reference according to the measured SOC; and (iii) a lower inner-loop control layer, which regulates the inverter's injected current and voltage to track the adjusted references. This layered structure keeps the SOC-awareness logic modular, so that it can later be extended or replaced (e.g., with a power-voltage variant) without altering the inner current-control loop.

3.4 Simulation and Test Scenarios

The following case studies, mirroring those used to validate the original strategy, will be simulated:

- **Case I – Fixed power references:** Both BESS-fed inverters operate at fixed $P_{ref} = 30$ kW and $Q_{ref} = 5$ kVAR, starting from different initial SOC values, to observe the active-power roll-off as SOC declines.
- **Case II – Step change in reactive power reference:** A step change in Q_{ref} is applied to one inverter to evaluate the dynamic response and decoupling between active and reactive power control.
- **Case III – Near-depleted battery:** Initial SOC values are set close to the lower protection limit to validate the automatic disconnection behaviour at $SOC = 35\%$.

- **P–V control comparison:** The SOC-based P–Q droop strategy is compared against a power–voltage (P–V) droop variant, in which the inverter regulates its output voltage directly rather than reactive power.

3.5 Performance Evaluation & Benchmarking

The simulated results (active and reactive power profiles, SOC trajectories, terminal voltage, and current waveforms) will be benchmarked against the results reported by Allabadi et al. [1] for the same test cases, to validate the correctness of the independent implementation. Performance will be evaluated in terms of: power-reference tracking accuracy, voltage regulation under the P–V strategy versus the P–Q strategy, and the effectiveness of the SOC-droop mechanism in preventing over-discharge while maintaining uninterrupted supply to the loads (via grid support).

Chapter 4

Results

4.1 Simulink Validation and SOC Based Control System

A validated MATLAB/Simulink model of the modified CIGRE low-voltage AC microgrid benchmark, incorporating two BESS-fed grid-connected inverters, PV, and wind-turbine sources. Figure 4.1 provides a basic overview of the system. Wind turbine and PV plant are included as a PV bus, and the grid is taken as a slack bus. Load are included as a series RLC load. Figure 4.2 gives the structure of the inverter implemented in this project. It has a simple LCL circuit with a SOC-based control. An independently implemented SOC-based droop control

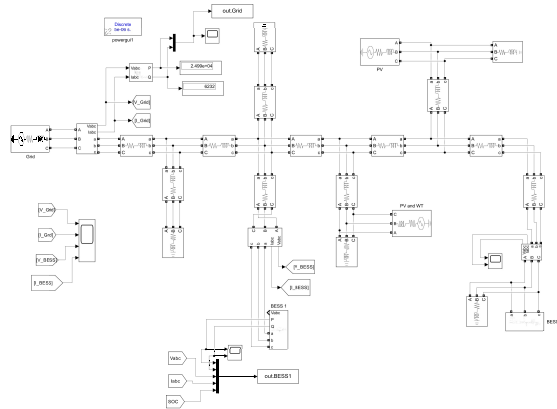


Figure 4.1: MATLAB Simulation of the CIGRE Model

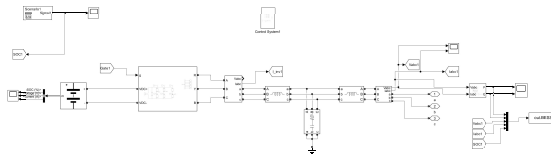


Figure 4.2: BESS included in the CIGRE Model

strategy for the BESS-fed inverter, including the outer power-control loop, inner current-control loop, and the SOC-voltage droop augmentation. The table ?? provides the values set for various components of the inverter.

Components	Value	Unit
Inductor	5.1m	H
Capacitor	95u	F
Capacitor(DC)	270u	F
Voltage line to line	400	V
Voltage(DC)	800	V
Frequency	50	Hz
Sampling Frequency	10k	Hz

Table 4.1: Components

4.2 Case-I to III

Simulation results for Cases I–III (fixed power reference, step change in reactive power reference, and near-depleted battery disconnection), benchmarked against the results reported in the reference study. Figure 4.3 shows that there is linear drop of the SOC level, this was created to analyze the system performance under various SOC in quick interval. The implemented battery was of 800V and 30Ah. To show the effectiveness of the system, implementation of the linear drop of SOC was performed.

4.2.1 Case I

Based on this Case-I can be observed in the figure 4.4. Since the SOC level in the BESS 1 was $> 85\%$. It was providing rated power, but the BESS 2 was already at 80% SOC it had a drop in the power injection. Similarly, when the SOC dropped below 85% in BESS 1, it then gradually dropped its power injection.

A constant reactive power is to be supplied in the system by the BESS, as observed in the figure 4.5. Reactive power injected to the grid was set to around

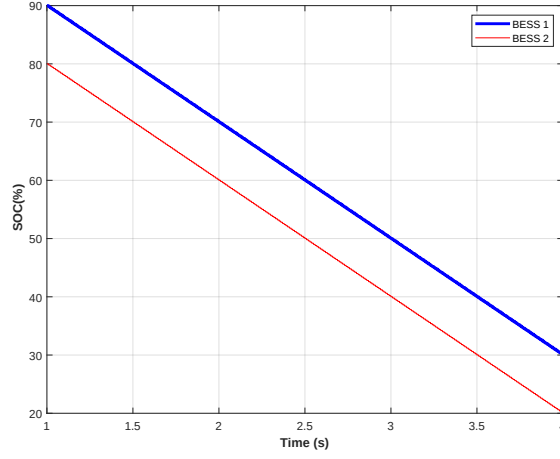


Figure 4.3: SOC drop Vs Time

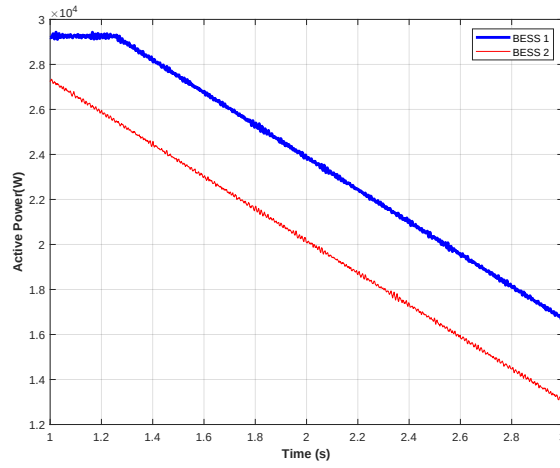


Figure 4.4: Drop in Active Power Vs Time

4.5kVAr. Finally, the observation 4.6 of the grid power injection shows that decrease in the BESS power injection increases the grid power.

4.2.2 Case-II

A step increase in reactive power is performed in the network. The constant reactive power is increased to 15kVAr from BESS 2. To perform the analysis regarding the effect of the injection of reactive power to the overall system. Figure 4.7 shows the injection of reactive power to the grid. Similarly, active power shows a transient during the injection of reactive power from the BESS to the grid as observed in the simulation result.

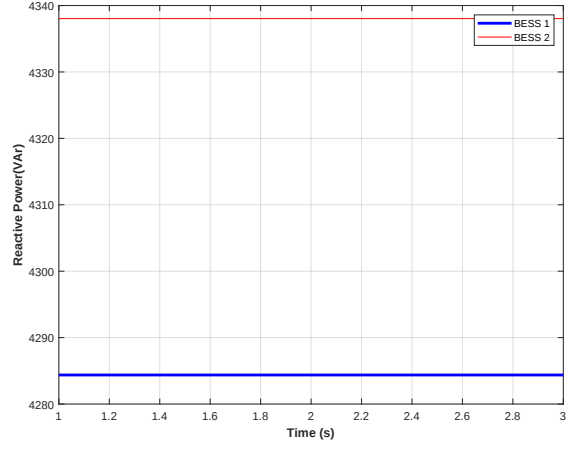


Figure 4.5: Reactive Power Injection

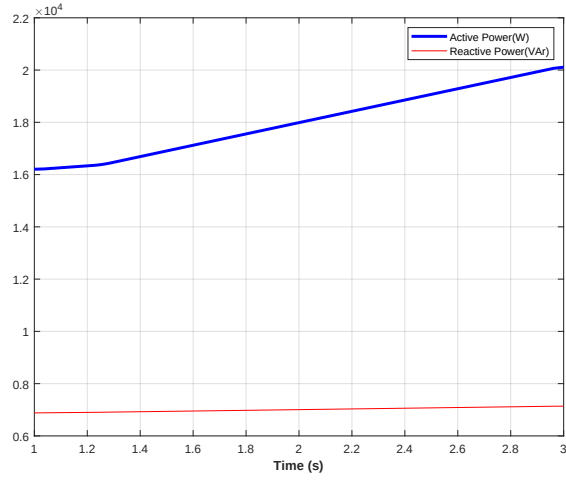


Figure 4.6: Increase in Grid Power

4.2.3 Case-III

The drop in the SOC $< 35\%$ should stop the inverter.

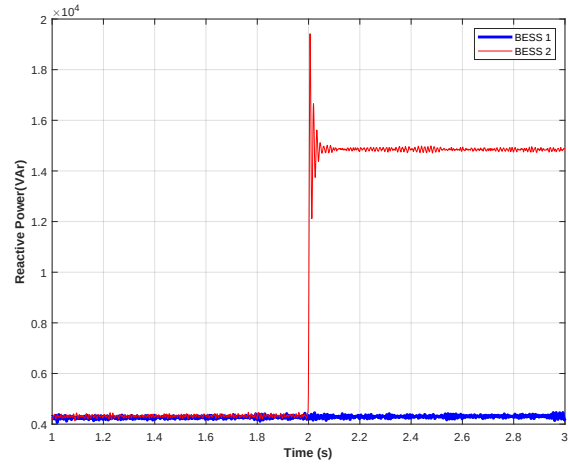


Figure 4.7: Step increase in Reactive Power

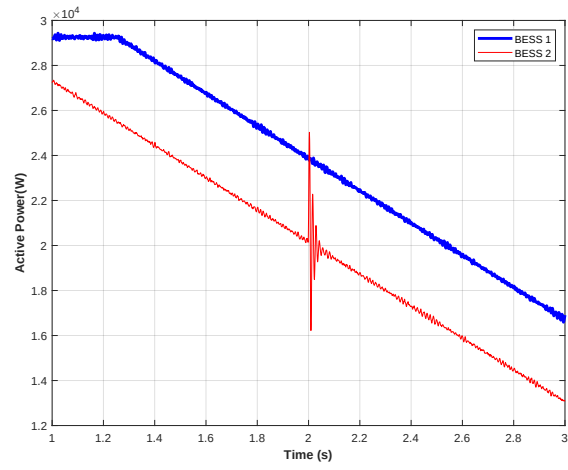


Figure 4.8: Effect of the reactive power on active power

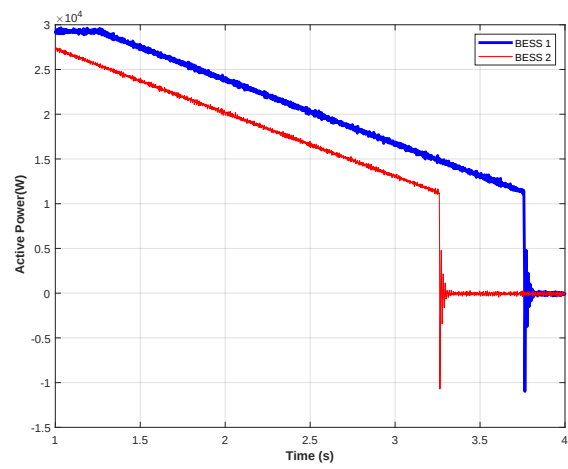


Figure 4.9: Control of Inverter after reaching 35% SOC

Chapter 5

Remaining Work

5.1 Expected Outcome from Remaining Work

- A comparative evaluation of the P–Q-based SOC droop strategy against the P–V droop variant, in terms of voltage regulation performance under declining SOC.
- A documented set of tuned controller parameters (PI gains, filter values, droop coefficients) suitable for further extension of the work, e.g., to multi-BESS coordination or additional SOC-balancing objectives.
- A dissertation/thesis chapter and, subject to results, a conference or journal submission summarising the validation study and any extensions identified during the work.

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